Huawei Access Node BAS-ACC-37 Routing Protocol Flap Resolution SOP

Node: BAS-ACC-37 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_3857

Description

This Standard Operating Procedure (SOP) outlines the resolution steps for a Routing Protocol Flap failure on Huawei Access Node BAS-ACC-37 at Basavanagudi POP. The failure is characterized by BGP route churn, OSPF adjacency loss, and ISIS LSP flooding, triggered by alarm code ALARM_3857 with a Critical severity. The underlying cause of this failure is attributed to a complex interplay between the routing protocol interactions and the Huawei device's hardware and software mechanics. Specifically, the Huawei device's BGP, OSPF, and ISIS protocols, which are integral to the routing protocol suite, can sometimes enter a recursive flap condition, leading to a rapid succession of route advertisements and withdrawals. This can cause the OSPF adjacency to flaps, leading to a loss of routing information and subsequently a routing protocol flap. Furthermore, the Huawei device's ISIS protocol, which is used for Level 2 routing, can also contribute to the routing protocol flap by flooding the network with LSPs, overwhelming the routing protocols and causing a flap. The combination of these factors can lead to a Critical failure, characterized by the ALARM_3857 alarm code.

Pre-requisite Checks

- Verify that the Basavanagudi POP is accessible and that the Huawei Access Node BAS-ACC-37 is powered on and operational.

- Confirm that the redundant link is operational and that the network is not experiencing any other Critical failures.

- Ensure that the Huawei device's configuration is up-to-date and that the latest software version is installed.

Potential Root Causes

Cause	Probability
BGP route churn caused by recursive flap condition	High
OSPF adjacency loss due to excessive routing protocol interactions	Medium
ISIS LSP flooding overwhelming the routing protocols	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp routing-table, Expected Output: BGP routing table information, including route prefixes and next-hop addresses, Description: This command displays the BGP routing table information, including route prefixes and next-hop addresses. This can help identify any BGP-related issues that may be contributing to the routing protocol flap.
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```
Command: display ospf neighbor, Expected Output: OSPF neighbor information, including neighbor IP addresses and state, Description: This command displays the OSPF neighbor information, including neighbor IP addresses and state. This can help identify any OSPF-related issues that may be contributing to the routing protocol flap.

Command: display isis lsp, Expected Output: ISIS LSP information, including LSP IDs and sequence numbers, Description: This command displays the ISIS LSP information, including LSP IDs and sequence numbers. This can help identify any ISIS-related issues that may be contributing to the routing protocol flap.
```

Resolution Steps

Step 1: Step: Check and clear BGP route churn, Command: clear bgp *, Description: This command clears the BGP routing table and restarts the BGP process. This can help resolve any BGP-related issues that may be contributing to the routing protocol flap.

Step 2: Step: Verify OSPF adjacency, Command: display ospf neighbor, Description: This command verifies the OSPF adjacency and checks for any issues that may be contributing to the routing protocol flap.

Step 3: Step: Check and clear ISIS LSP flooding, Command: clear isis lsp, Description: This command clears the ISIS LSPs and restarts the ISIS process. This can help resolve any ISIS-related issues that may be contributing to the routing protocol flap.

Post-Check Verification

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Command: display bgp routing-table, Expected Output: BGP routing table information, including route prefixes and next-hop addresses, Description: This command verifies that the BGP routing table is stable and that there are no BGP-related issues.

Command: display ospf neighbor, Expected Output: OSPF neighbor information, including neighbor IP addresses and state, Description: This command verifies that the OSPF adjacency is stable and that there are no OSPF-related issues.

Command: display isis lsp, Expected Output: ISIS LSP information, including LSP IDs and sequence numbers, Description: This command verifies that the ISIS LSPs are stable and that there are no ISIS-related issues.
```

Roll-back Procedures

- Step: Restore BGP configuration, Command: undo clear bgp *, Description: This command restores the BGP configuration and restarts the BGP process.

- Step: Restore OSPF configuration, Command: undo clear ospf, Description: This command restores the OSPF configuration and restarts the OSPF process.

- Step: Restore ISIS configuration, Command: undo clear isis, Description: This command restores the ISIS configuration and restarts the ISIS process.